

Hydrostatics

- Fluid vs. solid, pressure, units
- Eq. of equilib & soln. for incompressible & compressible
Examples - pressure gradient in water, mercury
isothermic atmosphere
- Application: hydraulics, hydraulic press & jack
- Buoyancy force
example: ship stability
- Problems
Pressure on dam (2025B)
Cone shaped container (2025A)

Hydrostatics

Fluid vs. solid:

Fluid at rest vs. solid
neutral equilib. stable equilib.

[Also: no long range order

Lattice]

* Later, distinguish incompressible & compressible fluids.
incompressible approximates liquids (as opposed to gases)
water air

Pressure

leading example

Fluid at rest can only support isotropic ^{negative} stress*

$$\boxed{d\vec{F} = -p d\vec{S}} \quad (*)$$

$p = p(\vec{x})$ same in all directions
pressure

* otherwise, in some frame $\exists$ non-diag (shearing) stresses - inconsistent w. static fluid.

No tension.

Comparing (*) w. $dF_i = \sigma_{ij} dS_j$

$$\Rightarrow \boxed{\sigma_{ij} = -p \delta_{ij}}$$

Dimensions & units

$$[p] = \frac{\text{Force}}{L^2} = \frac{\text{Energy}}{L^3}$$

MKS unit Pascal* - $1 \text{ Pa} := 1 \text{ N/m}^2$

* Blaise Pascal 1623 Clermont-Ferrand - 1662 Paris

French scientist (Mathematics, natural & applied science)

"clarified the concept of pressure and vacuum by

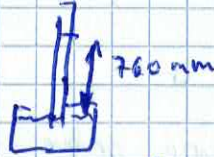
generalizing the work of Evangelista Torricelli"

Also: probability, ^{w. Fermat} work on projective geometry, pioneering work on mechanical calculator (1642) [later Leibniz 1672],

1st modern public transport (1660's). Philosophy & theology.

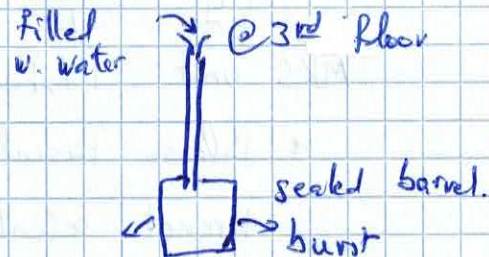
* Evangelista Torricelli 1608 Rome - 1647 Florence - see other side

- * Italian natural scientist, invented the barometer
Studied science in Rome under Castelli, a student of Galileo;
A suction pump can only raise water to ~10m. not taken
Torricelli's experiment (1643) filled a 1m long tube w. mercury



- Moved to Florence 3 months before Galileo's death^{on} 8 Jan 1642
Succeeded him as grand-ducal mathematician and
chair of Mathematics in U. of Pisa.
Published "Opera Geometrica" (1644) including cycloid
area & center of gravity.
"We live submerged at the bottom of an ocean of air"
("Noi viviamo sommersi nel fondo d'un pelego d'aria"
in a letter 11 Jun 1644).
Died in 1647 in Florence of fever, most likely Typhoid.

- * Pascal invented the hydraulic press & syringe.
Suggested & measured the decrease in pressure
with height.
Barrel experiment



Atmospheric pressure is known to fluctuate with the weather.

A typical value is

$$1 \frac{\text{kg-force}}{\text{cm}^2} = \frac{9.8 \text{ N}}{10^{-4} \text{ m}^2} \approx 10^5 \text{ Pa.}$$

Accordingly, one defines

$$- 1 \text{ bar} := 10^5 \text{ Pa}$$

$$- 1 \text{ standard atmosphere (Atm)} \approx 1.01 \text{ bar}$$

Additional units : mercury (Hg) millimeter

$$- 760 \text{ mm Hg} \equiv 1 \text{ Atm.}$$

Pound per square inch (PSI)

$$1 \text{ bar} \approx 10^5 \frac{\text{N}}{\text{m}^2} = \frac{10}{9} \frac{\text{kgf}}{\text{cm}^2} = \frac{10}{9} \frac{2.20 \text{ lbf}}{\text{in}^2} \left(\frac{1 \text{ in}}{2.54 \text{ cm}} \right)^2 = 14.5 \text{ PSI.}$$

Equation of hydrostatic equilibrium

Force per unit volume $0 = \partial_j \sigma_{ij} + \rho g_i = -\partial_j (p \delta_{ij}) + \rho g_i = -\partial_i p + \rho g_i$

$$0 = -\vec{\nabla} p + \rho \vec{g}$$

* Solution for incompressible fluids

$$\rho(x) = \rho_0 \quad \text{uniform } (\vec{x} - \text{indep.})$$

$$\Rightarrow \vec{g} = \vec{g}(\vec{x}) \text{ must be conservative (apply } \vec{\nabla} \times \text{ to eqn)}$$

$$\vec{g} = -\vec{\nabla} \phi, \quad \phi \text{ potential}$$

$$\Rightarrow \text{const} = -p(x) - \rho_0 \phi(x)$$

$$p(x) - p_0 = -\rho_0 \phi(x)$$

Most common case, uniform gravity

$$\boxed{p(x) - p_0 = \rho g h \quad \begin{array}{l} h - \text{depth} \\ p_0 = p(h=0), \rho = \rho_0 \end{array}}$$

Interp.: weight of liquid column
- Examples.

A. Water $\rho_0 = 1 \frac{\text{gr}}{\text{cm}^3} = 10^3 \text{ kg/m}^3$

Pressure gradient

$$\frac{\Delta p}{h} = \rho g \approx 10^3 \cdot 10 \text{ m/s}^2 = 10^4 \frac{\text{Pa}}{\text{m}} = 0.1 \frac{\text{bar}}{\text{m}} = \frac{1 \text{ bar}}{10 \text{ m}}$$

Known to divers.

B. Mercury (Hg) - a metal which is liquid at room temp.

$$\rho = 13.55 \text{ gr/cm}^3 \quad [Z \approx 80, A \approx 200.6, T_m = -38^\circ \text{C}]$$

$$\frac{h}{\Delta p} |_{1 \text{ atm}} = \frac{1}{\rho g} \times 1.01 \text{ bar} \approx 2.760 \text{ m} = 760 \text{ mm}$$

* Solution for compressible fluids.

Preparation - for this case we need some facts from thermodynamics & the theory of gases. For completeness we shall state them. For a more complete understanding, refer to the thermodynamics course (prerequisite).

We shall need - thermodynamic potentials

energy $E = E(S, V)^*$ $dE = TdS - PdV$

* The state of a gas is fully determined by 2 quantities

[free energy $F = F(T, V) = E - TS$ $dF = -SdT - PdV$

Gibbs Free energy $G = G(T, P) = E - TS + PV$ $dG = -SdT + VdP$

Enthalpy $H = H(S, P) = E + PV$ $dH = TdS + VdP$

We shall mostly use these quantities per unit mass.

- Ideal gas law

$$PV = nRT$$

$$= Nk_B T$$

n - number of moles
 $R = k_B N_A = 8.31 \text{ J/(mol} \cdot \text{K)}$

N - # of molecules

$$k_B = 1.38 \cdot 10^{-23} \text{ J/K} = 0.86 \cdot 10^{-4} \text{ eV/K}$$

It is an equation of state (relating the state variable P, V, T)

Equilibrium

Uniform temperature

$$T = T_0$$

no heat current

$$p = p(P, T_0) \quad \text{a func. of } P \text{ only.}$$

Solution. Equation 0 = $-\vec{\nabla} p - \rho(p) \vec{\nabla} \phi$.

Unknowns $p = p(\vec{x}), \rho = \rho(\vec{x})$
 1 diff. eq.
 1 alg. relation $\rho = \rho(p)$.

Define $\boxed{g^T(p) := \int_{\rho(p)}^p \frac{dp'}{\rho(p')}}$

(Denoted Φ in L&L)
 capital letter

Now, by chain rule $\vec{\nabla} g^T = \frac{dg^T}{dp} \vec{\nabla} p = \frac{1}{\rho(p)} \vec{\nabla} p$

Solution

$$\boxed{g^T(p(\vec{x})) + \phi(\vec{x}) = \text{const} = g^T(p(\vec{x}_0)) + \phi(\vec{x}_0)}$$

As before $\vec{g} = -\vec{\nabla} \phi$, ϕ potential

g^T is interpreted as Gibbs Free energy per unit mass.

Indeed

$$dG = -S dT + V dP$$

Fix T

$$dG = V dP$$

Per unit mass

$$g^T := \frac{SG}{Sm}$$

$$dg^T = \frac{SV}{Sm} dP = \frac{dP}{\rho} \checkmark$$

T - stands for thermodyn.
 to distinguish from accel.
 of gravity (when confusion
 might arise).

- Compressible fluid as special case:

$$g^T(p) = \int_{p_0}^p \frac{dp'}{\rho_0} = \frac{P}{\rho_0}$$

solution becomes $\frac{P}{\rho_0} + \phi(x) = \text{const} \checkmark$

Example: Isothermal atmosphere

What is the scale for atmosphere height?

What are the pressure and density profiles?

Composition of air

		(% by volume)	molar mass
N_2	78%		28.0
O_2	21%		32.0
H_2O	few %s		
Ar	1%		
CO_2	0.04%		

Eg. of state of uniform T^* (ideal gas law)

* this is an approx

$$PV = \text{const} \Rightarrow \rho \propto P, \rho = \frac{\rho_{STP}}{P_{STP}} P$$

At standard temp & pressure $p = 1 \text{ bar}$ $T = 0^\circ \text{C}$ (STP)

volume of 1 mole of ideal gas @ STP 22.4 L

mean molar mass of air 29.0 g/mol

(dominated by N_2).

$$\rho_0 = \frac{29.0 \text{ gr}}{22.4 \text{ L}} = \frac{29.0}{22.4} \cdot 10^{-3} \rho_w = \underline{1.29 \cdot 10^{-3} \rho_w} \quad (\text{dry air})$$

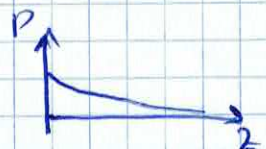
Equation & solution: $0 = -\vec{\nabla} p + \rho \vec{g}$

$$z \uparrow \Rightarrow \frac{dp}{dz} = -\rho g = -\frac{\rho_0 g}{p_0} \cdot p = -p/H$$

$$\text{Define } \left\{ H = \frac{p_0}{\rho_0 g} \quad \begin{matrix} p_0 = P_{STP} \\ \rho_0 = \rho_{STP} \end{matrix} \right.$$

$$\Rightarrow p(z) = p(z=0) \cdot \exp(-z/H)$$

$$\Rightarrow \rho(z) = \rho(z=0) \cdot \exp(-z/H)$$



Numerically $H = \frac{1 \text{ bar}}{1.29 \cdot 10^{-3} \rho_w \cdot 9.8} = \frac{10^5}{1.29 \cdot 9.8} = \underline{7.8 \text{ km}}$
 $\rho_w = 10^3 \text{ (kg/m}^3\text{)}$

Discussion

- H is typical length scale for the atmosphere
every H p, ρ drop by $1/e$.

compare w. height of tallest mountain (Everest 8,849m).

Half pressure height $H_{1/2} = H \log 2 = 5.4 \text{ km}$.

- We solved using the ideal gas law directly in equilib. eq.

Alternatively, we could determine g^T

$$g^T(p) = \int_{p(z=0)}^p \frac{dp}{\rho(p)} = \frac{p_0}{p_0} \int_{p(z=0)}^p \frac{dp}{p} = \left[\frac{p_0}{p_0} \log p/p(z=0) \right] = g^T(p)$$

Variant b) Gravitational field due to spherical mass w. M, r_0
neglect gas mass

$$g(r) = -\frac{GM}{r^2} = -g_0 \frac{r_0^2}{r^2} \quad g_0 = \frac{GM}{r_0^2}$$

$$\phi(r) = -g_0 r_0^2 / r$$

$$p(r) = p(r=r_0) \exp\left(\left[\phi(r_0) - \phi(r)\right] \frac{\rho_0}{p_0}\right)$$

$$p(r) = p(r=r_0) \exp\left[\left(\frac{1}{r} - \frac{1}{r_0}\right) \frac{r_0^2}{H}\right] \quad H = \frac{p_0 g_0}{\rho_0}$$

Variant c) Isothermal ^{self-gravitating} gas star (gravity due to gas).

$$\frac{dp}{dr} = -\rho g(r) = -\rho(r) \frac{GM(r)}{r^2} = -\frac{\rho(r)}{r^2} G \int_0^r 4\pi r'^2 \rho(r') dr'$$

$$\frac{p_0}{\rho_0} \frac{dp}{dr} =$$

$$\Rightarrow \left[\frac{d}{dr} \left(\frac{r^2}{\rho(r)} \frac{dp}{dr} \right) = -C r^2 \rho(r) \right] \quad C = 4\pi G \rho_0 / p_0$$

$$[C] = 1/m \checkmark$$

non-linear 2nd order diff. eq.

Hydraulics (application)

- * Pascal's principle: if height differences are negligible
 $\Delta h \ll H = \frac{p_0}{\rho g}$

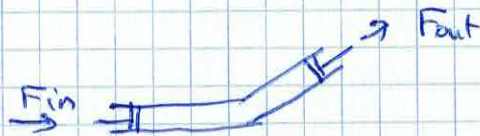
$$0 = \Delta p + \rho \Delta \phi \Rightarrow \Delta p = 0$$

$$\Rightarrow \boxed{p(x) = p_0 \quad \text{uniform pressure}}$$

- * Applied science / engineering involving mechanical properties of liquids.

Pascal's principle is used to manipulate forces:

- Displace & redirect force



pipe of uniform cross-section A , with 2 pistons (1/2 m) & seals (1/2 m)

$$p_{out} = p_{in} = \frac{F_{in}}{A} \Rightarrow |\vec{F}_{out}| = p_{out} A = |\vec{F}_{in}|$$

Force is translated & redirected with ease.
 Requires durable casing.

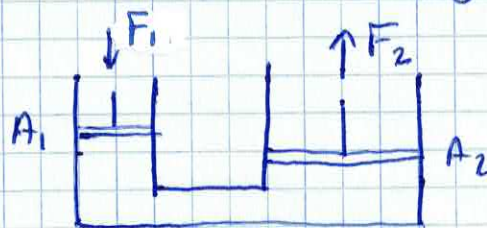
* Hydraulic press ~ a force multiplier.

Recall that force can be multiplied as long as the distance over which it acts is proportionally reduced so that conservation of energy holds

$$F_1 \Delta x_1 \geq F_2 \Delta x_2$$

$$\frac{F_2}{F_1} \leq \frac{\Delta x_1}{\Delta x_2}$$

The six simple machines (known since antiquity)
inclined plane, wedge, wheel, pulley, screw, lever.



$$F_2 = p A_2 \overset{\substack{\uparrow \\ \text{Pascal's principle}}}{=} \frac{F_1}{A_1} \cdot A_2 \Rightarrow \text{Force multiplier}$$

$$\boxed{\frac{F_2}{F_1} = \frac{A_2}{A_1}}$$

Conservation of energy is a result of volume conservation for incompressible fluid

$$F_1 \cdot \Delta x_1 = F_1 \cdot \frac{\Delta V}{A_1} = \frac{\Delta V}{P}$$

same for 1 $\rightarrow$ 2

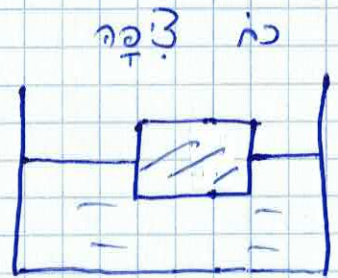
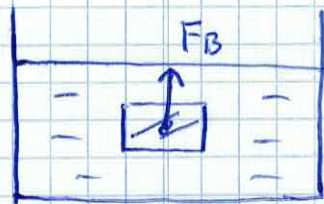
$$= F_2 \cdot \Delta x_2$$

See Youtube video: "How a hydraulic jack works"

Note valves & seals

PHYSICS PHYSICS

Buoyancy force



Consider a body submerged (fully or partially) by an incompressible fluid.

The body experiences an upward force.

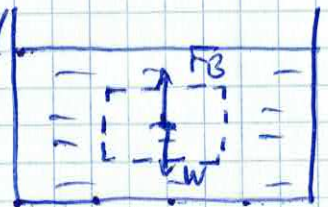
Why? $p(h_{\max}) > p(h_{\min})$.

What is the magnitude of F_B ?

Clever argument: Consider a container w. fluid only.

The pressure field remains identical

$$p'(\vec{r}) = p(\vec{r})$$



Since the fluid inside the dashed line is at equilibrium we must have

$$F_B = W_f = \rho_f V_f$$

V - displaced volume.
 ρ - density of displaced liquid.

In words: The buoyancy force equals the weight of the displaced liquid, and it acts through the center of volume.

Archimedes ~246 b.c.

Direct determination

$$\begin{aligned} -F_i &= \oint p dS_i = \int (p_0 + \rho g z) dS_i = \rho g \oint z dS_i = \rho g \oint z \delta_{ij} dS_j \\ &\stackrel{\text{div thm.}}{\Rightarrow} \rho g \int \partial_j (z \delta_{ij}) dV = \rho g \delta_{i3} \int dV = \rho g \Delta V \delta_{i3} \\ \vec{F} &= \rho g \Delta V (-\hat{z}) \end{aligned}$$

Ship stability and the drowning of the Vasa

(application)

The Vasa ship is a Swedish warship that made its maiden voyage on 10 Aug 1628 out of Stockholm harbor. At the time Sweden was a military regional power controlling areas in today's Sweden, Finland, and around the Baltic sea. The Vasa was built on the orders of king Gustav II Adolf, and was one of the most powerfully armed vessels in the world.

Some measures: 1210 tonnes displacement, total length $\sim 69\text{m}$, height 52.5m , propelled by sails, ^(1275m²) 64 guns (!), 145 sailors + 300 soldiers.

After sailing for 1300 m a gust of wind tilted it to the left, so much that the gun ports went under water, which rushed in and the ship sank to a depth of 32m, only 120m from shore.

30 persons were reported killed in the tragedy out of 100-200 on board. The ship sank in full view of a crowd, that assembled to watch the voyage.

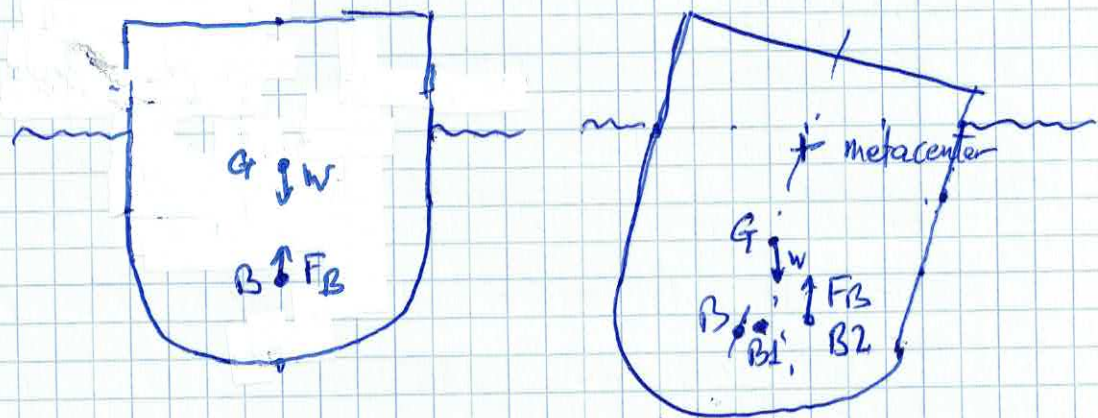
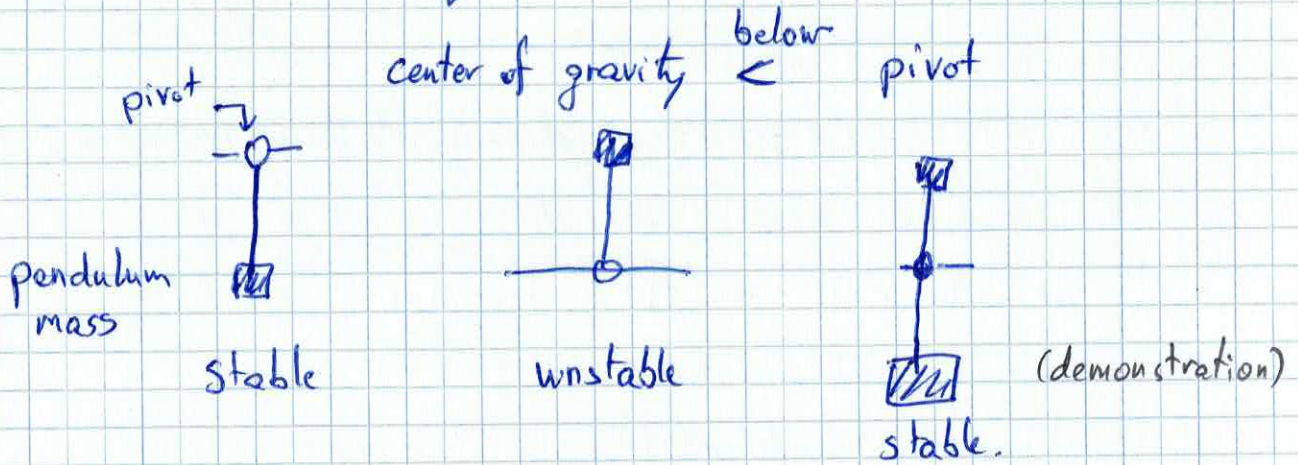
During the 1950's the ship was located, salvage in 1961 with a largely intact hull, and since 1990 is on display at the Vasa Museum in Stockholm, the most visited museum in Scandinavia.

Why did it sink? Due to poor understanding of mechanical stability.

[see virtual tour on Vasa Museum
→ explore → visit us from home website]

Ship stability

Recall that an object attached to a pivot is stable exactly if



The buoyancy force shift to lower ^{(side(right))} since it displaces more water.

If B_2 to right of $G \Rightarrow$ Righting moment, stable, metacenter above G

Metacenter = intersection of F_B line with center line.

Conversely if B_1 to left of G : unstable destabilizing moment.